**Use: Amylase and pH: Continuous-Sampling Investigation**

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

## STAFF ANSWERS · Supported

### Arrival

Amylase is the enzyme and speeds up starch breakdown.

Yes, starch is detected in the sampled drop.

Starch fits amylase's active site.

Prepare the iodine wells before mixing and starting timing.

### Exit

90 s; starch is not detected in that sample.

Record 90 s. We did not observe 75 s; the endpoint happened after 60 s and by 90 s.

### Supported independent task

1. 90 s; starch is not detected in that sampled drop.

2. Prepare and check → start together → sample at fixed times → keep the raw observation. Recording occurs as sampling proceeds; the last card names the evidence habit throughout.

3. Temperature stays matched. pH is the variable changed; matching temperature helps isolate the pH comparison.

4. First observed endpoint = 90 s. The exact moment between 60 s and 90 s is unknown.

5. pH 7 is faster in this series: 90 s compared with 240 s at pH 4, under the stated matched conditions.

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## STAFF ANSWERS · Independent

### Arrival

Amylase is the enzyme; starch is the substrate.

90 s. The exact change between the two samples was not observed.

Independent: buffer pH. This is what we change. Dependent: endpoint time. This is what we measure. A valid control is temperature. It could be the solution volume or concentration. It could be the sampling interval.

Get ready first. Then you can take samples and make observations at the agreed times. You are less likely to miss what happens early on.

### Exit

Record 90 s; the endpoint was reached after 60 s and by 90 s.

For example, keep temperature matched so a temperature change cannot explain a difference attributed to pH.

### Independent independent task

1. First observed endpoint = 90 s; it was reached after 60 s and by 90 s. Blue-black at 60 s shows starch still detected then.

2. Prepare/check → start → sample → record honestly as sampling proceeds. Preparation avoids losing observations while the reaction continues.

3. IV: buffer pH. DV: endpoint time. Keep temperature or starting amounts/volumes/concentrations matched, so their changes do not offer another explanation.

4. Record 90 s as the first observed endpoint, not an invented 75 s. Record the missed sample explicitly and retain actual later observations.

5. For example pH 7: 90 s, versus pH 4: 240 s, so pH 7 is faster for these conditions. One trial at each of five pH values cannot identify an exact universal optimum.

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## STAFF ANSWERS · Aspire

### Arrival

The colour shows if starch is left. If it disappears when conditions are controlled, this supports breakdown. It does not directly measure what the molecules do.

No. Record 90 s as the first observed endpoint; the endpoint was reached after 60 s and by 90 s.

Match temperature across the pH trials. Then a change in temperature is less likely to explain the results. Compare with no active amylase. This checks for change when the enzyme does not act.

No. If 30 s was positive and 90 s negative, the observed window is wider: after 30 s and by 90 s. Preserve the missed sample.

### Exit

Still record first observed endpoint = 90 s; the window is after 30 s and by 90 s, widening from 30 s to 60 s.

A shorter fixed interval, if authorised and feasible, narrows the timing window. It does not remove subjective judgement of borderline iodine colour.

### Aspire independent task

1. Record first observed endpoint = 90 s. The endpoint was reached after 60 s and by 90 s, a 30 s window; its exact moment is unknown.

2. Preparation prevents a delay at the start; fixed intervals give comparable sampling opportunities and a stated timing limit. Neither removes all colour-judgement uncertainty.

3. Matched temperature prevents a competing temperature effect across the pH trials. The no-active-amylase comparison checks whether starch disappearance requires active enzyme under the same conditions.

4. The endpoint was reached after 30 s and by 90 s, a 60 s window. Record 60 s as missed and the first observed endpoint as 90 s; the timing precision is reduced.

5. The pH 7 endpoint is earlier (90 s versus 240 s) under matched conditions. This first series supports faster breakdown at pH 7 than pH 4 here. Raw observations must remain visible; uncertainty calls for checking or an authorised repeat, not a preferred value.

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## STAFF ANSWERS · shared tasks and mark scheme

### Shared activities

Starter · ready before mixing: Prepare labelled iodine wells and the recording times before starting the authorised reaction. The reaction does not pause while equipment is prepared. Sampled-drop model: 30 s and 60 s: blue-black, starch detected. 90 s: orange-brown, starch not detected. Record first observed endpoint 90 s; it was reached after 60 s and by 90 s. Method-decision checkpoint: Continue as authorised: ready checked tile; clean fixed sample. Stop/check adult: iodine spill; unlabelled buffer; temperature outside the authorised range. Preserve evidence and seek an authorised repeat/check: missed interval; borderline colour. Never invent a missing observation. Record and repair: Replace “75 s exactly” with “first observed endpoint 90 s; reached after 60 s and by 90 s”. A missed 60 s sample widens the known interval from after 30 s to by 90 s. Keep the missed result visible.

### Independent

1. State the first observed endpoint for pH 7 and the interval in which it was reached, using the supplied observations. [2 marks]

Record 90 s (1 mark)

The crossing was after 60 and by 90 s (1 mark)

### Independent

2. Order the four method cards. Explain why preparation must happen before mixing. [2 marks]

Prepare/check, start together, sample at fixed times, record raw observations (1 mark)

Reaction starts on mixing, so preparation must be completed first (1 mark)

### Independent

3. State the independent variable, dependent variable and one matched control, explaining why the control matters. [4 marks]

Change pH (1 mark)

Measure endpoint time (1 mark)

Keep temperature or solution amounts matched (1 mark)

The named control could affect rate independently of pH (1 mark)

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## STAFF ANSWERS · shared tasks and mark scheme · continued

**Independent**

4. Repair the proposed exact endpoint of 75 s. Explain what should be recorded if a sample is missed. [2 marks]

75 s was not observed, so record 90 s (1 mark)

Keep a missed-sample gap and use actual observed bounds (1 mark)

**Independent**

5. Compare two pH values using their times. State one limit of the conclusion from this first series. [3 marks]

For example pH 7 takes 90 s and pH 4 takes 240 s (1 mark)

pH 7 is faster in these matched first runs (1 mark)

Single trials, tested range or subjective endpoint limits the claim (1 mark)

**Aspire**

1. For pH 7, give the endpoint time we record. Show how this differs from the true endpoint window. State how wide the window is. [3 marks]

Record 90 s (1 mark)

Threshold crossing is after 60 and by 90 s (1 mark)

The window is 30 s wide (1 mark)

**Aspire**

2. Explain why we get the tile ready first. Explain why we take samples at a fixed time gap. Show how these protect different parts of the evidence. [2 marks]

Preparing the tile avoids losing early observations (1 mark)

A fixed sampling interval gives consistent timing gaps (1 mark)

**Aspire**

3. Explain why we keep temperature the same. Explain why we use a comparison with no active amylase. Show how their roles differ. [2 marks]

Matching temperature limits its effect as an alternative cause (1 mark)

The no-active-amylase comparison tests change without active enzyme (1 mark)

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## STAFF ANSWERS · shared tasks and mark scheme · continued

**Aspire**

4. Evaluate this claim: The 60 s sample is missing in a second record; 30 s is blue-black and 90 s orange-brown. State what is now known and repair the claim “the same precision”. [3 marks]

The observed crossing is after 30 and by 90 s (1 mark)

The window is 60 s wide (1 mark)

It is wider than the original 30 s window (1 mark)

**Aspire**

5. Compare pH 4 and pH 7 with the times given. Keep your claim to these trials. You might expect a smooth trend. A colour might be hard to read. Explain why neither reason lets you change a raw observation. [3 marks]

pH 7 has a shorter endpoint time than pH 4: 90 versus 240 s (1 mark)

The conclusion is limited to these matched trials (1 mark)

Changing observations to fit an expected trend would make the record untruthful (1 mark)

**Aspire**

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.

**Choice**

Use the matching independent-task answers; preserve the pupil’s own scientific sentence, label or reason. Staff mark I, S or C beside this task.